Arista Access Node ELEC-ACC-01 Routing Protocol Flap SOP - Electronic City Hub

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_4247

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve routing protocol flap issues on the Arista Access Node ELEC-ACC-01 at the Electronic City Hub. Routing protocol flaps occur when the BGP route churn, OSPF adjacency loss, or ISIS LSP flooding on the node cause instability in the network, leading to critical alarm ALARM_4247. This is due to the inherent design of the Arista Access Node, which relies on the efficient exchange of routing information between nodes to ensure optimal routing decisions. The Arista EOS (Extensible Operating System) implements the BGP, OSPF, and ISIS routing protocols, which are susceptible to flapping due to various reasons such as misconfiguration, hardware or software issues, or even external network changes. The node's role as an access point in the Electronic City Hub makes it crucial to quickly identify and resolve any issues to prevent network downtime and data loss.

Pre-requisite Checks

- Verify that the redundant link status is functioning properly.

- Ensure that the remote hands are dispatched to the Electronic City Hub to perform physical checks and troubleshooting.

- Confirm that the node's power and cooling systems are functioning within normal operating parameters.

Potential Root Causes

Cause	Probability
BGP route churn due to misconfigured neighbor relationships	High
OSPF adjacency loss due to hardware or software issues	Medium
ISIS LSP flooding due to external network changes	Low

Diagnostic Commands & Expected Outputs

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Command: show bgp summary, Expected Output: BGP table version 2, main routing table version 2, Description: This command displays a summary of the BGP process, including the number of peers, routes, and messages. It helps identify any issues with the BGP neighbor relationships, which can lead to route churn.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays information about the OSPF neighbors, including their state and dead time. It helps identify any issues with OSPF adjacency, which can lead
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to adjacency loss.

Command: show isis database, Expected Output: ISIS LSP database, Description: This command displays the ISIS LSP database, which includes information about the LSPs and their attributes. It helps identify any issues with the ISIS routing protocol, which can lead to LSP flooding.
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Resolution Steps

Step 1: Step: Verify and correct BGP neighbor relationships, Command: show bgp summary | include neighbor, Description: Verify that the BGP neighbor relationships are correctly configured and that there are no issues with the BGP process. Correct any misconfigurations or issues found during the diagnostic process.

Step 2: Step: Reset OSPF adjacency, Command: clear ospf process, Description: Reset the OSPF process to reestablish adjacency with the neighbors. This step helps resolve issues with OSPF adjacency loss.

Step 3: Step: Clear ISIS LSP database, Command: clear isis database, Description: Clear the ISIS LSP database to resolve issues with LSP flooding. This step helps ensure that the ISIS routing protocol is functioning correctly.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP table version 2, main routing table version 2, Description: Verify that the BGP process is functioning correctly and that there are no issues with the BGP neighbor relationships.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: Verify that the OSPF neighbors are functioning correctly and that there are no issues with OSPF adjacency.

Command: show isis database, Expected Output: ISIS LSP database, Description: Verify that the ISIS LSP database is functioning correctly and that there are no issues with the ISIS routing protocol.
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Roll-back Procedures

- Step: Restore previous configuration, Command: copy startup-config running-config, Description: Restore the previous configuration to ensure that the node returns to its original state.

- Step: Reset OSPF adjacency, Command: clear ospf process, Description: Reset the OSPF process to reestablish adjacency with the neighbors. This step helps resolve issues with OSPF adjacency loss.

- Step: Clear ISIS LSP database, Command: clear isis database, Description: Clear the ISIS LSP database to resolve issues with LSP flooding. This step helps ensure that the ISIS routing protocol is functioning correctly.